RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Thursday, 6 August 2026, 10:18 AM
State	Finished
Completed on	Thursday, 20 August 2026, 11:39 AM
Time taken	14 days 1 hour
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int count = 0;

void function(int n)
{
  int i, s;

  i = 1;
  count++;

  s = 1;
  count++;

  while (1)
  {
    count++;

    if (!(s <= n))

      break;

    i++;
    count++;

    s += i;
    count++;
  }
}

int main()
{
  int n;
  scanf("%d", &n);

  function(n);

  printf("%d", count);
```

```
40
41      return 0;
42  }
43
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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